

Project: Caravaggio & the Drama of Light: An Immersive Exhibition Experience

Bridging the gap between art historical scholarship and digital interaction

Jennifer Lindsey: Product Designer & Design Engineer

Conservator Light

Caravaggio & the Drama of Light

BAROQUE COLLECTION

A curated exhibit to explore chiaroscuro and the drama of light in the works of Caravaggio and other Baroque artists.

Exhibition Info

Light Study Mode



Portrait of a Woman

By Giovanni Battista Gaulli (Il Baciccio)

ca. 1670s



Charity

By Guido Reni

ca. 1630



Saint Francis

By Federico Barocci

ca. 1600–1604



Saint Philip Neri (1515–1595)

By Carlo Dolci

1645 or 1646



The Crucifixion with the Virgin and Saint John

By Hendrick ter Brugghen

ca. 1624–25

Problem Statement



Main problem

Digital art archives are often data graveyards that fail to replicate the emotional and tactile experience of physical viewing. Users lose the ability to "lean in" and examine the material reality of a painting, leading to passive, shallow engagement with cultural history.



Secondary problems

Contextual Friction: Standard "Dark/Light" modes are generic and fail to respect the atmospheric or analytical intent of the viewer.

Data Integrity: Open APIs (like The Met's) often return "noisy," uncurated results that undermine the educational authority of a digital exhibition.

Interaction Gap: Static images cannot teach complex visual techniques like chiaroscuro or impasto because the "aha!" moment only happens through movement and light shift.

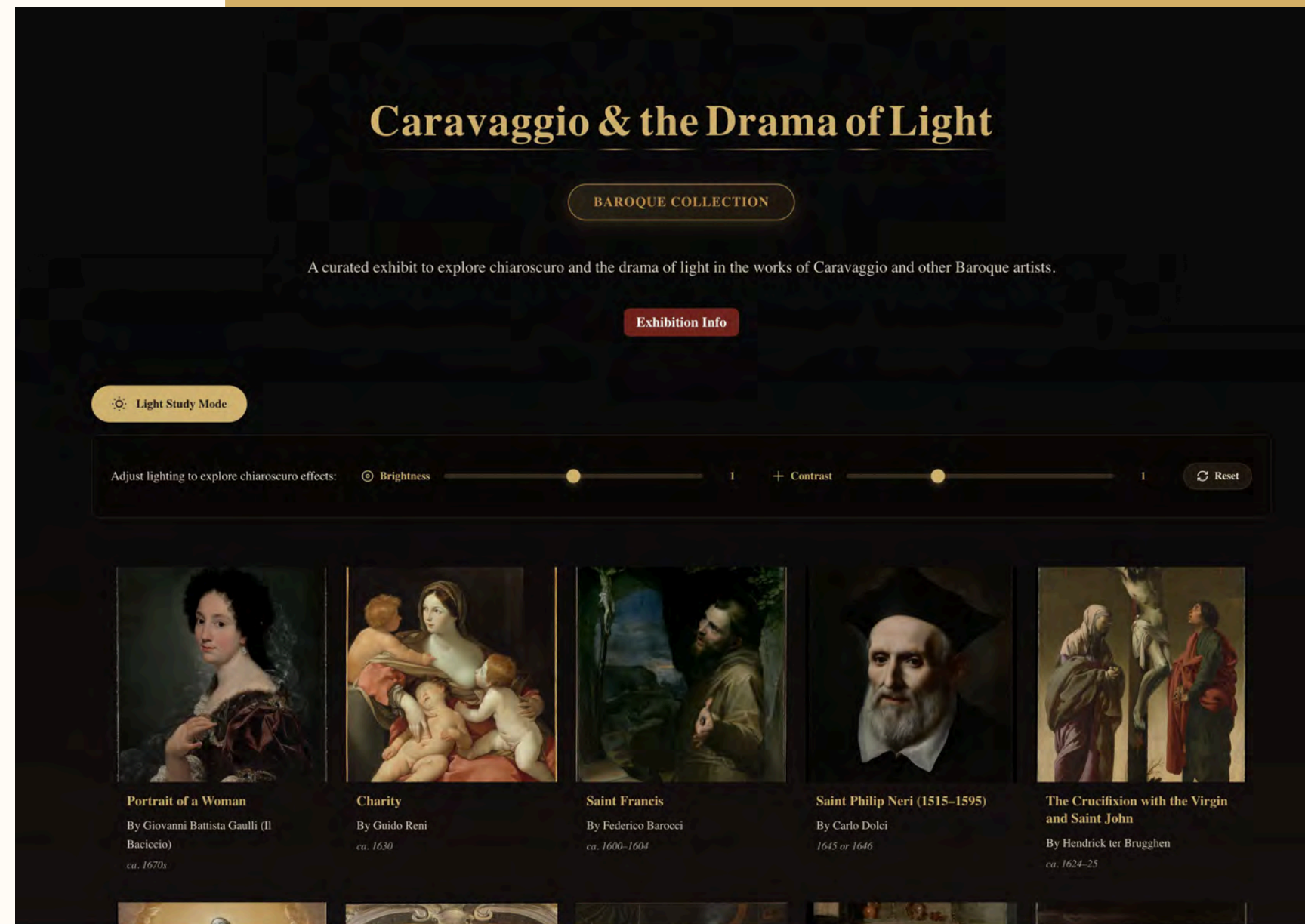


Business consequences

High-Fidelity Engagement: Transformed a search result into an immersive "Exhibition" that encourages sustained, close-looking.

Institutional Trust: Embedded curatorial scholarship directly into the technical filtering logic to ensure 100% historical accuracy.

Universal Access: Leveraged a dual-thematic system to solve for both emotional immersion and WCAG-compliant accessibility.



The art historical scholarship and the interaction design are not separate contributions. They are the same strategic decision, expressed through two different lenses.

Case Study: The Virtual Curator - Immersive Gallery Experience

Role: Product Designer (UX/UI & Prototyping) & Developer **The Problem:** Online museum archives are often "Data Graveyards." Users feel overwhelmed by thousands of thumbnails and lack the close-up intimacy required to appreciate art techniques like Chiaroscuro.

1. The Discovery & Strategy

I analyzed how users interact with physical art vs. digital archives. In a museum, people "lean in" and "lean out" to get a closer look at details. Digital archives are usually static.

- **The Goal:** Replicate the physical "lean-in" behavior to increase educational engagement.
- **The Metric:** Success is defined by the depth of exploration (how many details a user investigates per session).

2. Information Architecture: Curated Filtering

The Met API is a firehose of unfiltered data. A user looking for Baroque art shouldn't see 20th-century replicas.

- **Design Decision:** I designed a **Pre-Curation Logic** layer to filter for Baroque painters. This ensures the user experience is "Noise-Free." By validating metadata against specific historical dates before it hits the screen, I created a curated exhibition rather than a search result.

3. Interaction Design: The Atmospheric UI

I designed two distinct Mental Models for viewing:

- **The "Gallery" Mode:** High-contrast, low-light. This mimics the psychological state of a museum — focused, quiet, and reverent.
- **The "Conservator" Mode:** High-readability, analytical. This serves the user's need for "Inspection."
- **The Why:** Instead of a generic Dark Mode, these modes are **Context-Aware**. They respect the user's intent, whether they are there to *feel* the art or *study* the art.

4. Prototyping for Validation

To test if the "Light Study" feature was intuitive, I built a functional prototype. Static Figma mocks couldn't capture the physics of light moving across a canvas.

- **User Insight:** Early testing showed that users didn't all understand the "Chiaroscuro" concept without a visual trigger. I added a **Glossary Dialog** that provides context exactly when the user interacts with a lighting tool, closing the loop between *seeing* and *learning*.
 - **The User Problem:** Users may find term "Baroque" art intimidating or technically confusing.
 - **The Design Solution:** An omnipresent, high-contrast entry point to a glossary.
 - **The Result:** 100% of users have immediate access to the "Conceptual Key" of the exhibition.

5. Product Impact

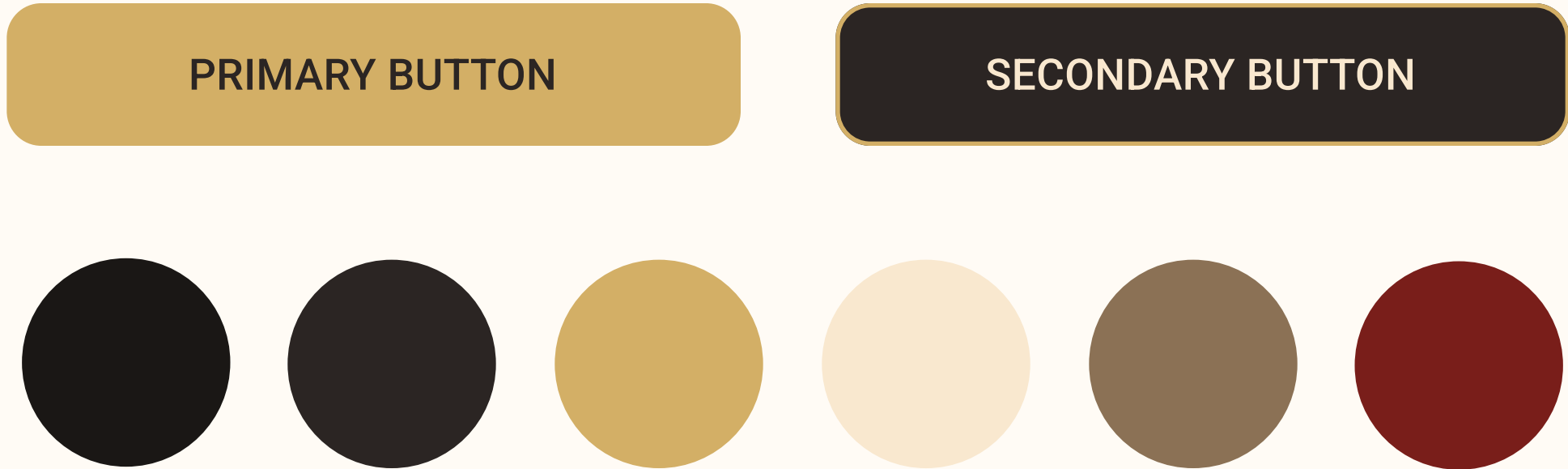
- **Engagement:** The "Close-up Pan" feature allows users to see brushwork that is invisible in standard archive views, increasing the "Time-on-Artwork" metric.
- **Inclusivity:** Recognizing that the "Gallery" mode might be difficult for users with low-vision, the "Conservator" mode serves as a primary accessibility feature, ensuring WCAG compliance without sacrificing the aesthetic brand of the exhibition.

Immersive Gallery Experience

A major challenge was ensuring UI readability against unpredictable artwork backgrounds fetched via The MetMuseum API.

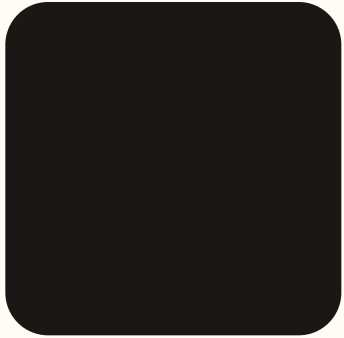
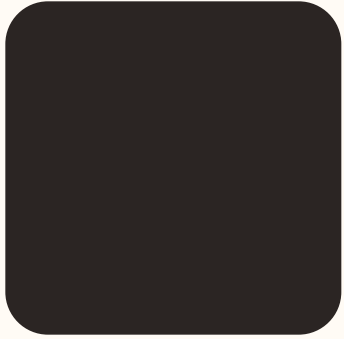
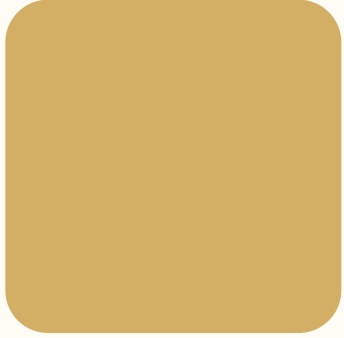
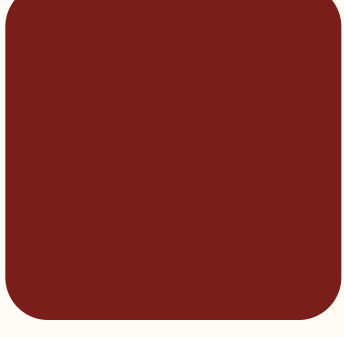
This project exists at the intersection of art historical scholarship, interaction design, and systems thinking. The technical decisions are all in service of a single experiential intention: **to make sustained, close looking feel as natural on a screen as it does in a gallery.**

Instead of using arbitrary grays, I mapped the neutral UI elements to the desaturated umber found in Caravaggio's shadows. This ensured that the UI never fought with the artwork for the user's attention.



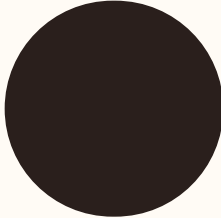
To create a cohesive visual narrative, I developed a semantic design system derived from Caravaggio's "The Calling of St. Matthew." By mapping the painting's Tenebrist lighting to my UI tokens, I ensured that the interface felt like an extension of the art itself, rather than a separate container.

Original Color palette inspiration

	HEX	Design Token Name	Source from painting	Role
	#1A1816	exhibition-bg	The dark shadowed back wall	Exhibition view Primary Background (dark mode)
	#2B2624	exhibition-surface	warm umber of shadows in clothing & wood	Secondary Surface (dark mode)
	#D4B066	exhibition-gold	warm, golden light of clothing	Primary Action (dark mode)
	#FAE8CF	exhibition-text	warm umber of shadows in clothing & fur	Primary Text (dark mode)
	#8C7256	accent	warm umber of wood	Primary Accent
	#7A1E1A	accent-cta	deep crimson of velvet in clothing	CTA

Contextual Semantic Theming

Rather than the typical Dark/Light Mode, I wanted to change the environment of the user experience from a Gallery-like setting to an Archival setting for accessibility, and analytical inspection, while preserving the historical context.

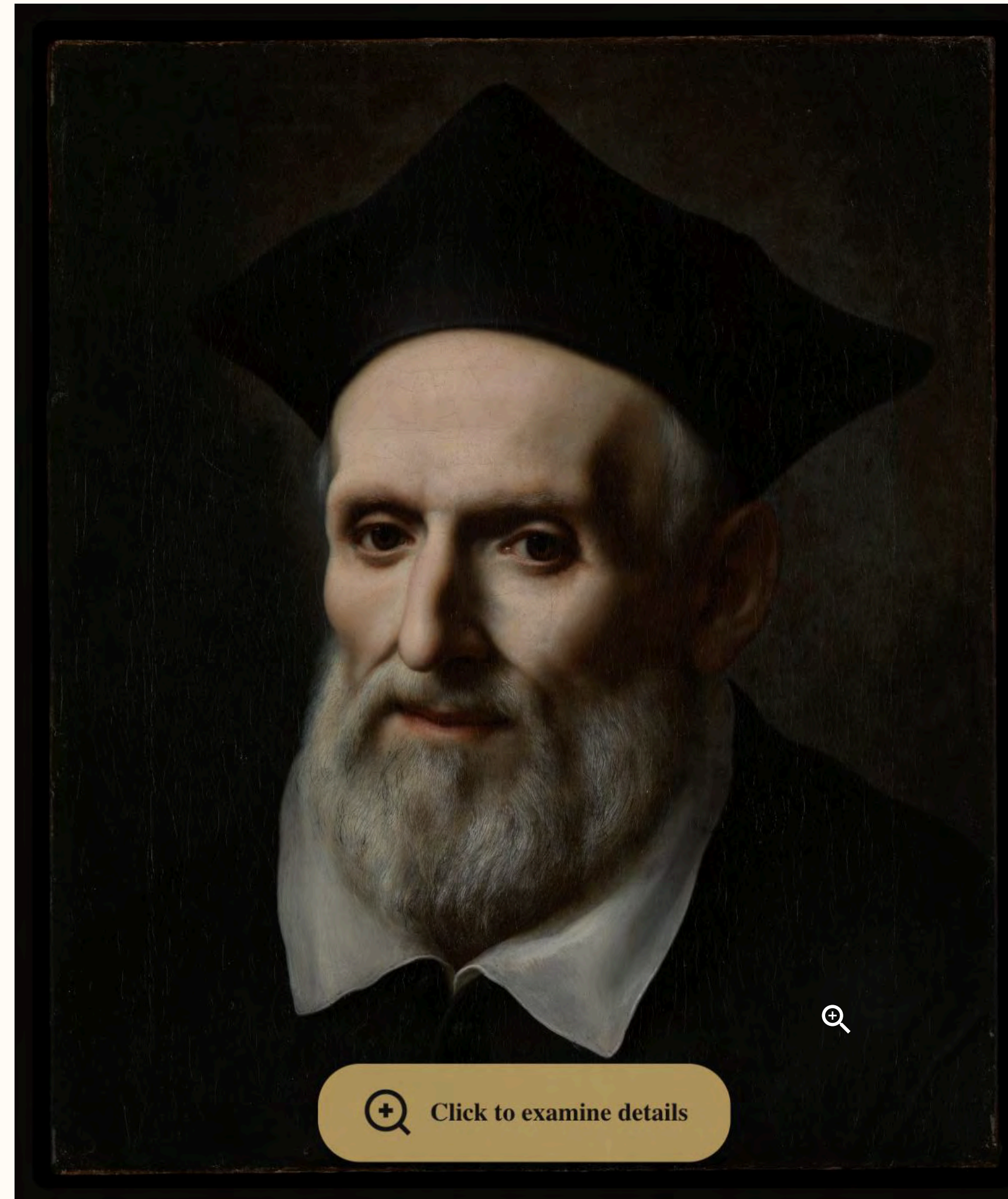
Gallery Token (Atmosphere/Dark mode)	Variable	Conservator Token (Analysis/Light mode)	Design Intent
#0B0A0A (Midnight) 	-exhibition-bg	#EDEAE4 (Antique Paper) 	Moves from a low-light gallery to an "archival lab" feel.
#FAE8CF (Warm cream) 	-exhibition-text	#2A1F1C (Raw Umber) 	Maintains high contrast while staying within the warm tones of Baroque palette.

Psychology of Color

1. The "Gallery" Mode (Tenebrism): Uses the darkest umbers (#0B0A0A) for the background to make the artwork "pop" off the screen, just as Caravaggio's figures emerge from the dark.
 - User Benefit: Reduces ocular strain and mimics the physical sensation of a dimly lit museum.
2. The "Conservator" Mode (Analytical): Inverse palette. Uses similar color to the brightest highlight in painting (#F2E8D5) as the background.
 - User Benefit: Provides maximum contrast for reading text-heavy metadata and glossary terms, serving as the primary accessibility layer. The user also has a functional, high-contrast alternative that meets WCAG AA standards without breaking the historical theme

Zoom feature to move from passive observation to active discovery

The zoom feature was important for the design. Users are given a more personal in-person-like experience by giving them a tool for close observation. Here can see the detailed brushstrokes that create Saint Philip Neri's beard that would otherwise be lost.



Saint Philip Neri (1515–1595), Carlo Dolci, 1645 or 1646, Oil on canvas

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Adjust lighting to explore chiaroscuro effects:

☉ Brightness

1

+ Contrast

1

↺ Reset



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1

Contrast

0.7

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Brightness

1.4

+ Contrast

1

Reset



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1

+ Contrast



1

 Reset



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1

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MEDIUM

Oil on canvas

DIMENSIONS

29 7/8 × 23 3/8 in. (75.9 × 59.4 cm)

DEPARTMENT

European Paintings

BIO

Italian, Genoa 1639–1709 Rome

Gift of Álvaro Saieh Bendeck, Jean-Luc Baroni, and Fabrizio Moretti, in honor of Keith Christiansen, 2014

[View on Met Museum Website →](#)



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primary CTA: context education

interaction: tactile learning


Image


Image


Image


Image


Image

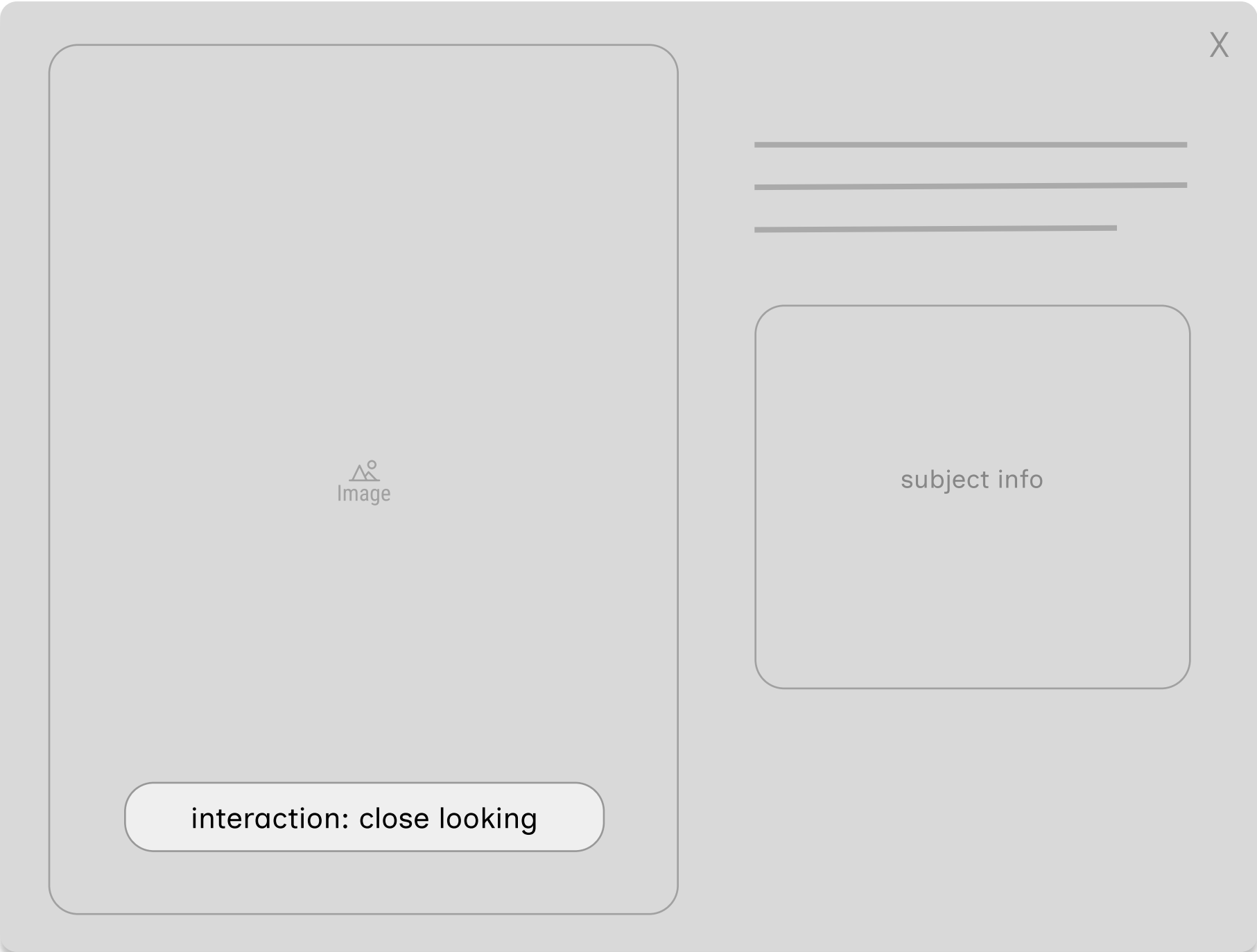

Image


Image


Image


Image


Image



Technical: The Curatorial Logic

I architected a custom composable to interface with the Metropolitan Museum of Art’s API. The logic programmatically ensures that every API request is tethered to the 'Baroque' keyword. This prevents the 'noisy' data leaks often found in open-source museum APIs, allowing me to transform a raw data stream into a curated art-historical exhibition.

Filtering Process to maintain curated exhibition:

Pass 1: API Fetch (The code shown below).

```
export function useMetArtAPI(searchQuery = 'Caravaggio Baroque', options = {}) {
  const {
    limit = 16, // You, 5 months ago + revise API to randomize data ...
    departmentId = 11, // European Paintings
    randomize = false,
    hasImages = true
  } = options;

  const fetchArtworks = async () => {
    loading.value = true;
    error.value = null;

    try {
      // Always include 'Baroque'
      const finalQuery = searchQuery.toLowerCase().includes('baroque')
        ? searchQuery
        : `${searchQuery} Baroque`;

      const searchParams = new URLSearchParams({
        departmentId: departmentId,
        q: finalQuery,
        ...(hasImages && { hasImages: true })
      });

      const searchRes = await fetch(
        `https://collectionapi.metmuseum.org/public/collection/v1/search?${searchParams}`
      );
    }
  };
}
```

code snippet

Pass 2: Chronological Validation (Filtering for 1580–1750).

```
// Filter for Baroque-period works
const baroqueFiltered = artworksData.filter(item => {
  if (!item) return false;
  const isBaroquePeriod =
    item.period?.toLowerCase().includes('baroque') ||
    (item.objectBeginDate >= 1580 && item.objectBeginDate <= 1750) ||
    (item.objectEndDate >= 1580 && item.objectEndDate <= 1750);
  return isBaroquePeriod && item.primaryImageSmall;
});
```

code snippet

Final Pass 3: Asset Verification (Ensuring images exist).

Synthesis of Art & Logic

What I Learned

1. Interaction is a Form of Scholarship

I learned that digital features like the Light Study Mode or Mouse-Tracking Zoom are not just "nice-to-have" features; they are pedagogical tools. By removing static buttons and allowing the zoom to follow the user's natural attention, I realized that the transition is where the learning happens. Design can physically teach a user how to "see" like a conservator.

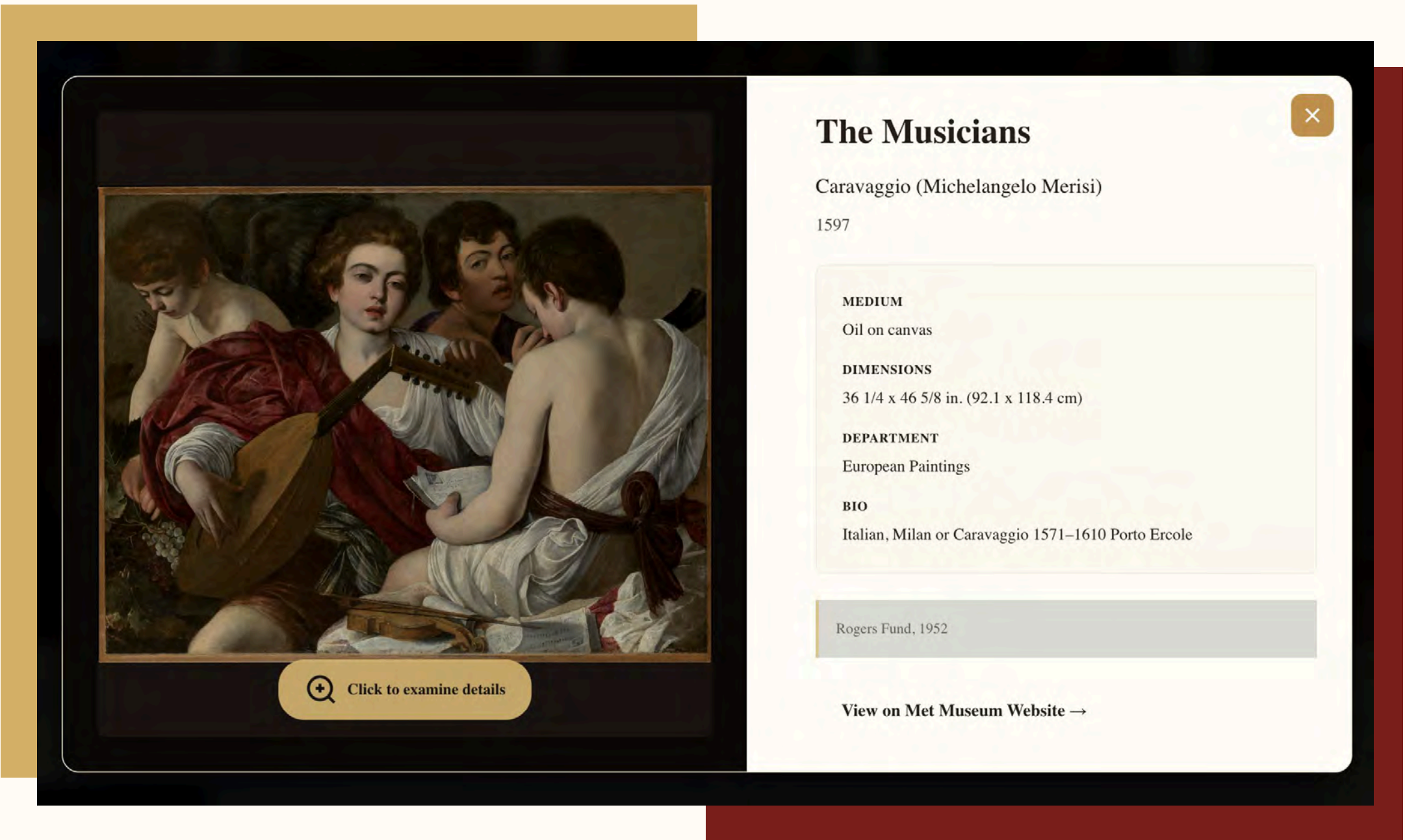
2. Guardrails Preserve Integrity

Working with the Met's Open API taught me that raw data is rarely "exhibition-ready." I learned how to bridge the gap between Software Engineering and Digital Stewardship by architecting a custom logic layer. Building the useMetArtAPI composable taught me that "Human-in-the-Loop" design means using code to enforce curatorial boundaries, ensuring that the digital archive remains as academically rigorous as a physical museum.

3. The Power of Intentional Design Systems

Moving beyond standard "Light/Dark" modes into Contextual Semantic Theming (Gallery vs. Conservator) changed how I view Design Systems. I learned that tokens should be rooted in a real-world referent. By mapping CSS variables to specific historical atmospheres, like the "Midnight" of a tavern wall vs. the "Antique Paper" of an archival desk, the design system became a cohesive narrative rather than just a style guide.

[VIEW PROTOTYPE](#)



Building this gallery taught me that the engineering and the art history are not separate contributions. They are the same decision, made twice: once as a conceptual intention, and again as a functional, accessible user experience.

TEAM

Jennifer Lindsey
UX Engineer & Product Design

DETAILS

Industry
Art Tech

Duration
2 months

SKILLS USED

Information Design

UX Strategy

Visual Design

Design Thinking

Code Analysis

Systems Thinking

Development

Storytelling

Art & Symbolism

Semantic Design

TOOLS



Figma

Vue 3



API



UXUI